

Annex A: Calculation of circular angle spread

For a signal with N multi-paths, each has M sub-paths, the conventional angle spread calculation for the composite signal is given by

$$\sigma_{AS} = \sqrt{\frac{\sum_{n=1}^N \sum_{m=1}^M (\theta_{n,m,\mu})^2 \cdot P_{n,m}}{\sum_{n=1}^N \sum_{m=1}^M P_{n,m}}} \quad (\text{A-1})$$

where $P_{n,m}$ is the power for the m th subpath of the n th path, $\theta_{n,m,\mu}$ is defined as

$$\theta_{n,m,\mu} = \text{mod}(\theta_{n,m} - \mu_{\theta} + \pi, 2\pi) - \pi, \quad (\text{A-2})$$

μ_{θ} is defined as

$$\mu_{\theta} = \frac{\sum_{n=1}^N \sum_{m=1}^M \theta_{n,m} \cdot P_{n,m}}{\sum_{n=1}^N \sum_{m=1}^M P_{n,m}} \quad (\text{A-3})$$

and $\theta_{n,m}$ is the AoA (or AoD) of the m th subpath of the n th path. Note that we have dropped the AoA (AoD) subscript for convenience.

We note that the angle spread should be independent of a linear shift in the AoAs. In other words, by replacing $\theta_{n,m}$ with $\theta_{n,m} + \Delta$, the angle spread $\sigma_{AS}(\Delta)$ which is now a function of Δ should actually be constant no matter what Δ is. However, due to the ambiguity of the modulo 2π operation, this may not be the case. Therefore the angle spread should be the minimum of $\sigma_{AS}(\Delta)$ over all Δ :

$$\sigma_{AS} = \min_{\Delta} \sigma_{AS}(\Delta) = \sqrt{\frac{\sum_{n=1}^N \sum_{m=1}^M (\theta_{n,m,\mu}(\Delta))^2 \cdot P_{n,m}}{\sum_{n=1}^N \sum_{m=1}^M P_{n,m}}} \quad (\text{A-4})$$

where $\theta_{n,m,\mu}(\Delta)$ is defined as

$$\theta_{n,m,\mu}(\Delta) = \text{mod}(\theta_{n,m}(\Delta) - \mu_{\theta}(\Delta) + \pi, 2\pi) - \pi, \quad (\text{A-5})$$

$\mu_{\theta}(\Delta)$ is defined as

$$\mu_{\theta}(\Delta) = \frac{\sum_{n=1}^N \sum_{m=1}^M \theta_{n,m}(\Delta) \cdot P_{n,m}}{\sum_{n=1}^N \sum_{m=1}^M P_{n,m}} \quad (\text{A-6})$$

and $\theta_{n,m}(\Delta) = \text{mod}(\theta_{n,m} + \Delta + \pi, 2\pi) - \pi$.